Ex Undis, Utilitas

Utility from a Wave of Quantum Information

Timothy C. Burt

2025-12-01

1. Ad Astra
   - An Application by Any Other Name
2. Quantum Information Science and Technology
   - Waves Front
3. Quantum Very Long Baseline Interferometry
   - The Starlight is Going Through a Phase
4. Phase Closure
5. Support Material
   - References

A Night Sky in Grand Teton National Park



Orion at pre-dawn (Tim Burt, Nikon D7000, AF-S Nikkor 20mm, F1.8, 8s)

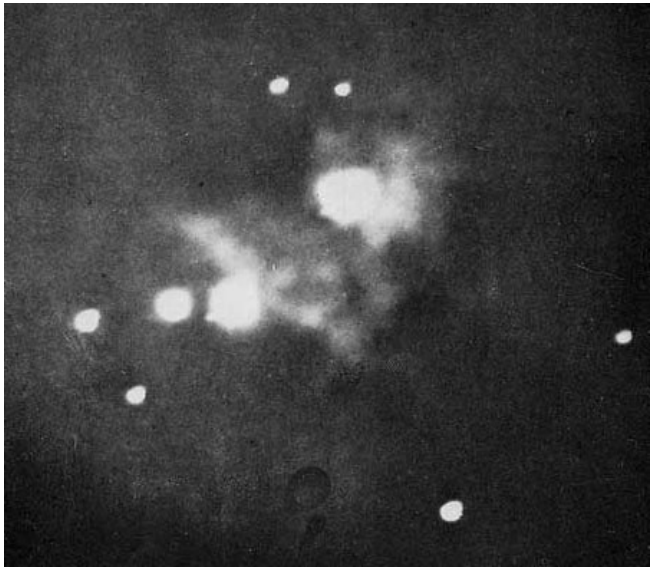
When I Heard the Learn'd Astronomer

When I heard the learn'd astronomer,
When the proofs, the figures, were ranged in columns before me,
When I was shown the charts and diagrams, to add, divide, and measure them,
When I sitting heard the astronomer where he lectured with much applause in the lecture-room,

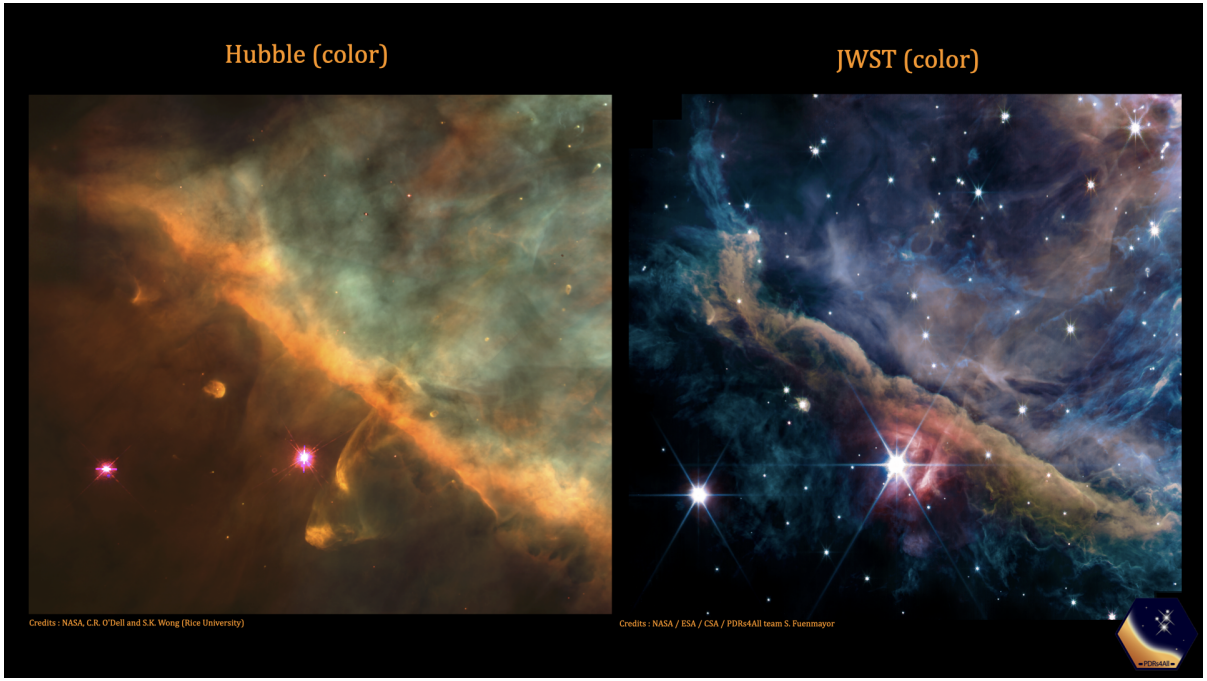
How soon unaccountable I became tired and sick,
Till rising and gliding out I wander'd off by myself,
In the mystical moist night-air, and from time to time,
Look'd up in perfect silence at the stars.

— Walt Whitman, 1865

Orion Through the Years and Better Eyes



First image of Orion, taken by Henry and Anna Draper, 1880 [5]

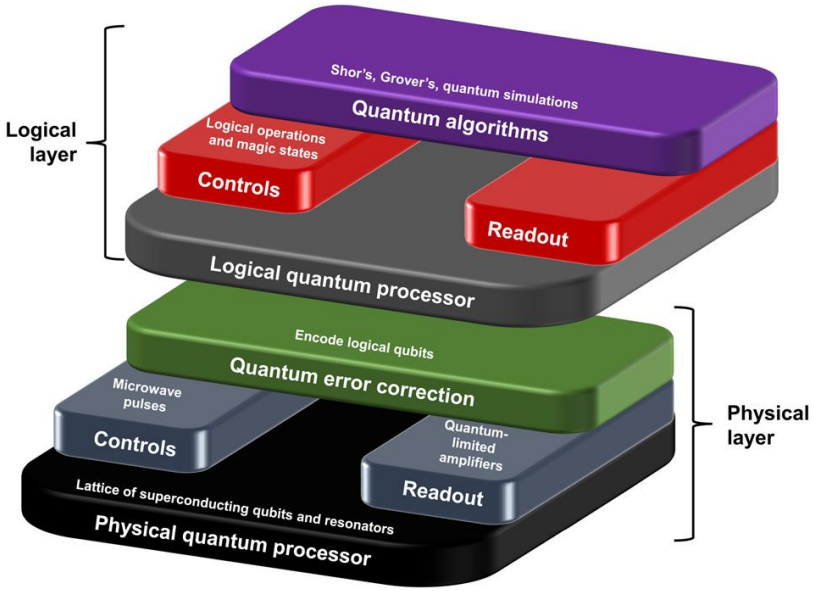


Orion from Hubble Space Telescope and James Webb Space Telescope [6]

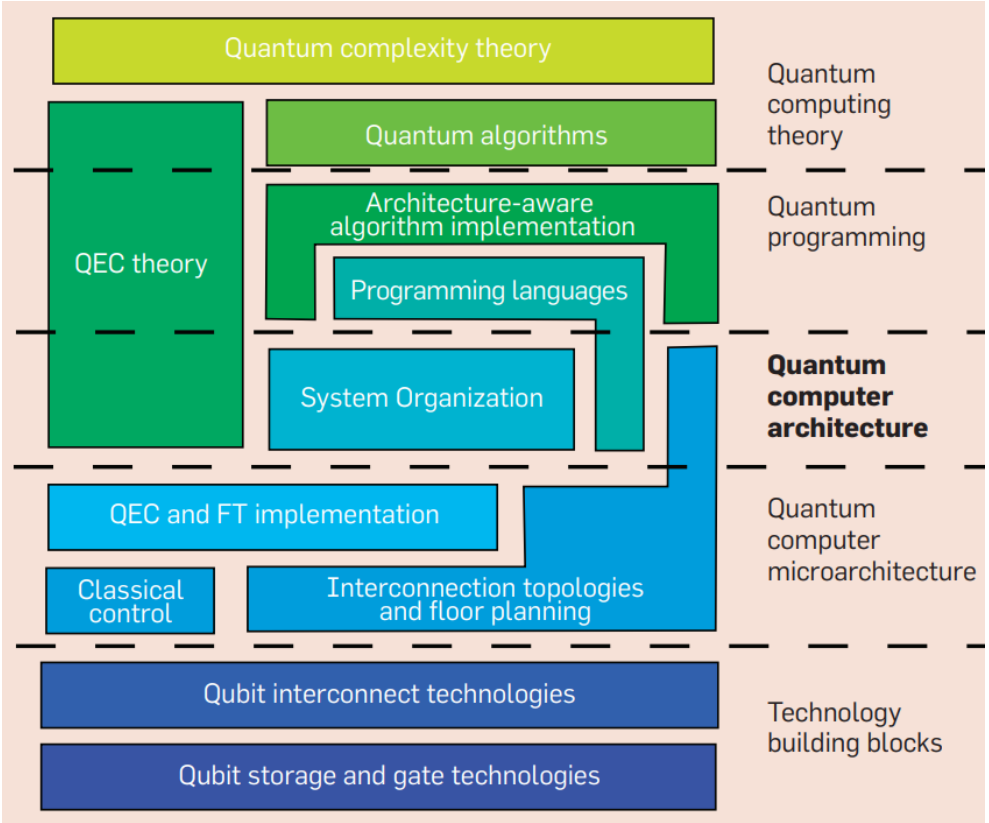
Electromagnetic waves from roughly 1,500 light years away of the Orion Nebula (M24) are captured on film and digital detectors.

The never-ending question: What is it good for?

Quantum Computing Stacks



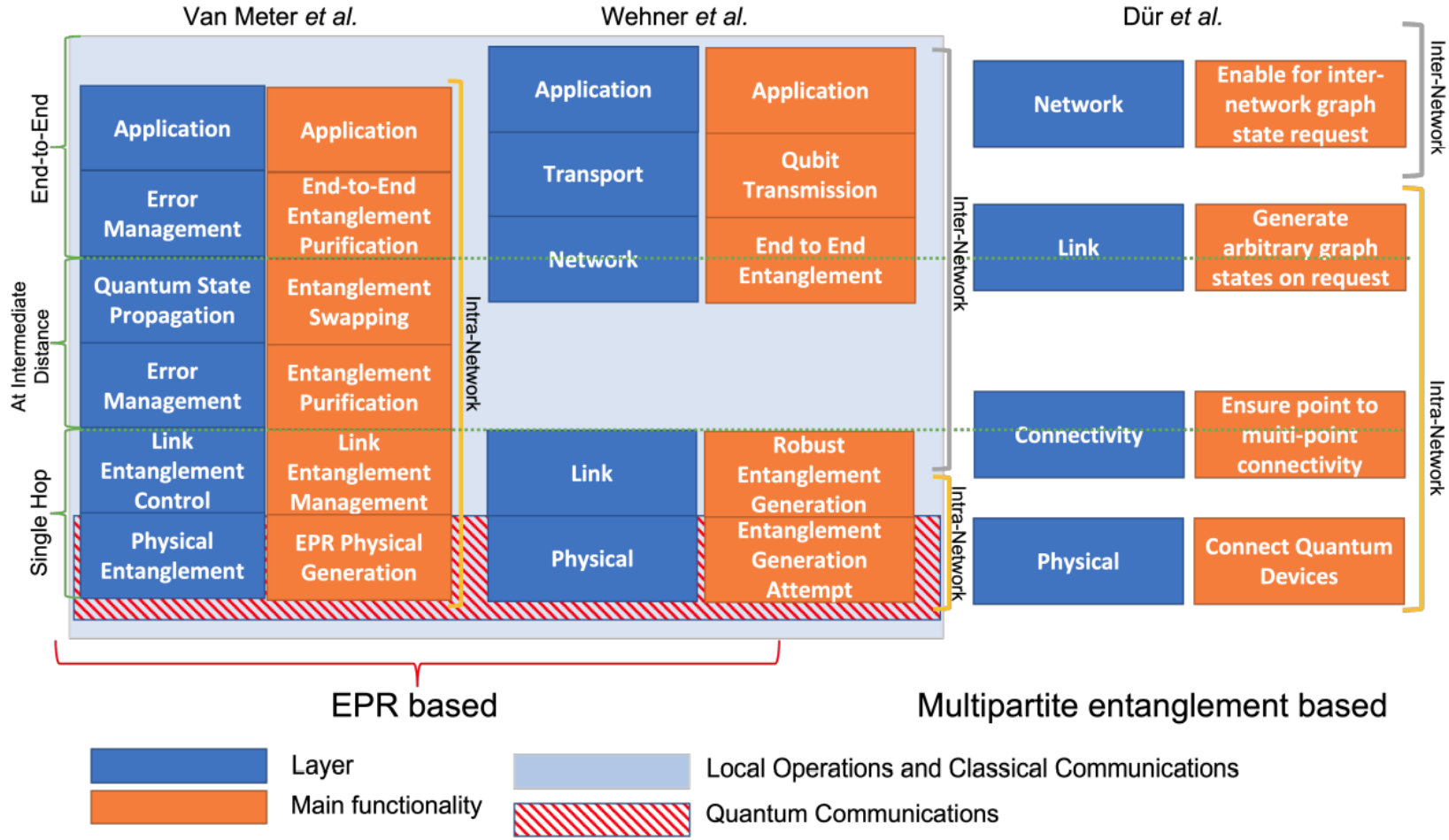
A notional stack for quantum computers from IBM [15]



A more detailed view of quantum computer architecture from the Association for Computing Machinery [14]

.

Quantum Networking Stacks



Comparison of three quantum network stacks [4]

Applications of Quantum Information

Sensing

- spatial measurements (e.g. translation, rotation)
- metrology of physical quantities (e.g. electric field, gravity)
- imaging (e.g. ghost imaging, computational imaging)

Computing

- factor prime numbers (e.g. crack encryption)
- database search
- optimization (e.g. traveling salesman problem)
- machine learning

Communication

- provably-secure key generation and distribution (e.g. cryptography)
- network security using untrusted nodes
- transaction protection (e.g. financial, voting)
- collective exposure (e.g. secret sharing)
- non-repudiation (e.g. contracts)

Simulation

- chemical reactions
- genetics
- pharmaceuticals

Enablers to Objectives

Quid nostra refert? (Of what importance is it to us?)

Enablers

- single photon light source
- photon number resolving detector
- low-loss classical component
- photonic integrated circuit

Applications

- quantum key distribution
- quantum secret sharing
- distributed quantum computing
- distributed quantum sensing

Technologies

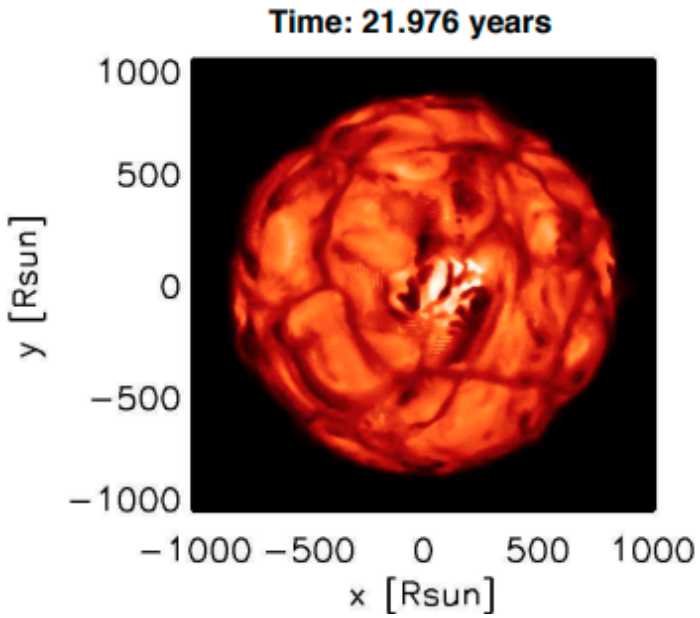
- quantum memory
- transduction
- quantum repeater
- quantum processor

Objectives

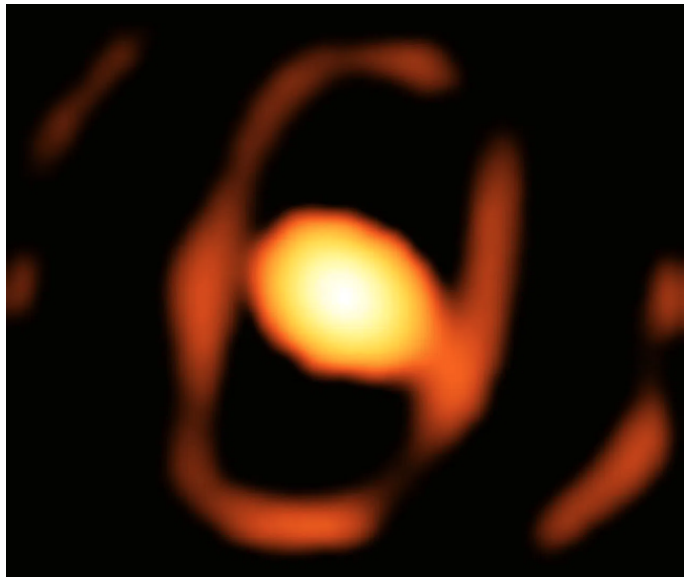
- contract non-repudiation
- chemical interaction modeling
- alternative GPS
- astrophysical imaging

Astrophysical Imaging

Ex astris, scientia (From the stars, knowledge)—Starfleet Academy motto



Simulation of red super giant [7]



"First-ever close-up of a star [WOH G64] in another galaxy" [8]

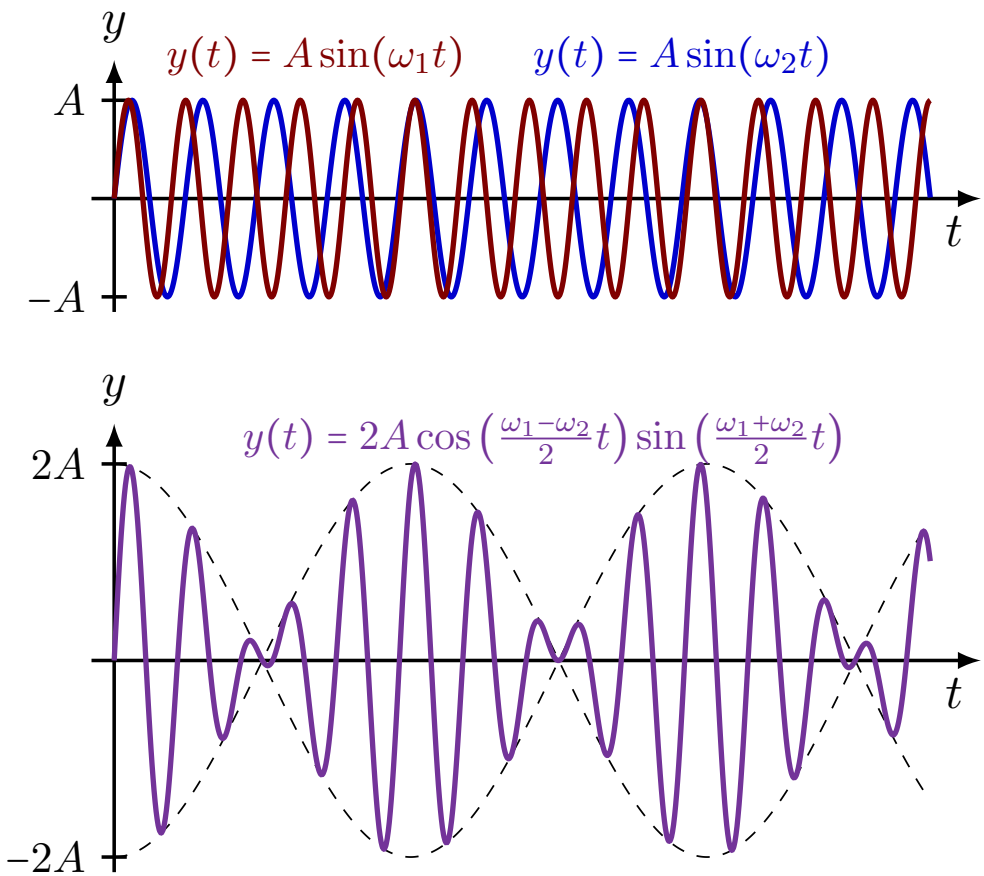
The simulation was used in the 2010 Interferometric Imaging Beauty Contest [9] to represent Gamma Crucis (γ Cru) at ~ 89 light years away. The WOH G64 close-up is from actual *interferometric imaging* of the star at 1.6×10^5 light years away.

An electromagnetic wave equation

$$\frac{\partial}{\partial t}u(\mathbf{r},t) = c^2\nabla^2 u(\mathbf{r},t)$$

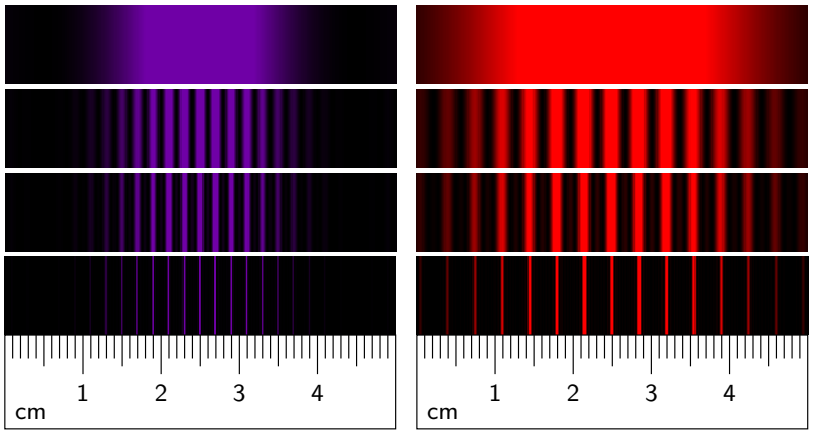
Simple Wave Interference: Different Frequencies

Waves add such that the resultant amplitude can be greater than the amplitude of either (constructive interference) or less than (destructive interference), depending on the relative phase.

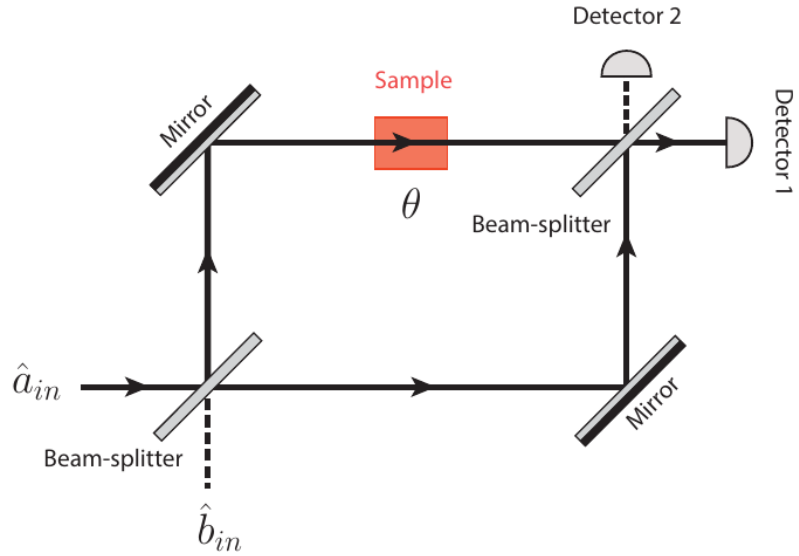


Interference from Slits and Splitters

"[I]nterference is always associated with the intrinsic indistinguishability of [. . .] paths" [13]



Interference patterns from 1, 2, 3, and 9 slits (from top to bottom) for $\lambda = 400\,\text{nm}$ (left) and $\lambda = 700\,\text{nm}$ (right) input to slits of width $10\,\mu\text{m}$ separated by $10\,\text{mm}$ at a distance of $0.5\,\text{m}$ from the detection plane



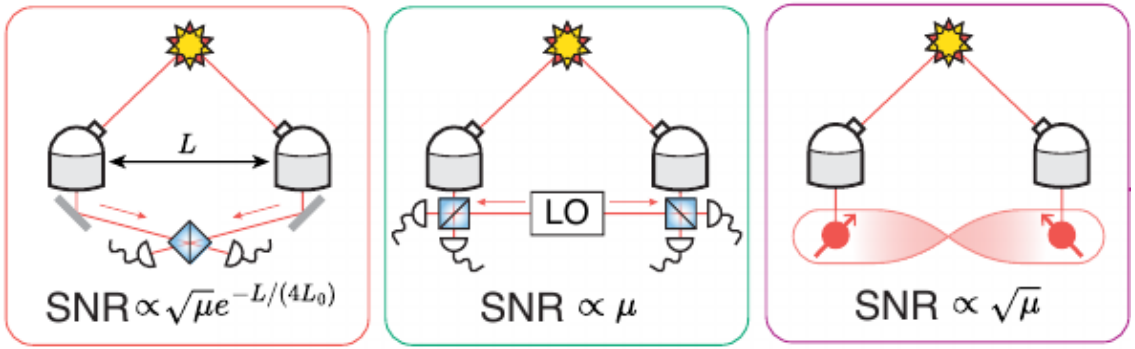
Two-port Mach-Zehnder interferometer [12]

An Introduction of Interferometric Imaging

Physical insight The complex degree of coherence at two points $j(\vec{r}_1, \vec{r}_2; t)$ from an *incoherent source* increases via propagation (van Cittert-Zernike theorem); see animated visual aid on Wikipedia

Physical connection In the far zone, $j(\vec{r}_1, \vec{r}_2; t)$ is proportional to "the normalized Fourier transform of the **intensity distribution across the source**" [10]

Physical constraint Phase knowledge degrades with distance for classical direct detection, limiting the separation L (current $L_{\text{max}} \approx 600\,\text{m}$), thereby limiting angular resolution $\alpha_{\text{res}} \propto \lambda / L$



Comparison of signal to noise for three interference schemes—direct detection, phase reference, and quantum-assisted—for low photon numbers $\mu \ll 1$ (weak signals) [11]

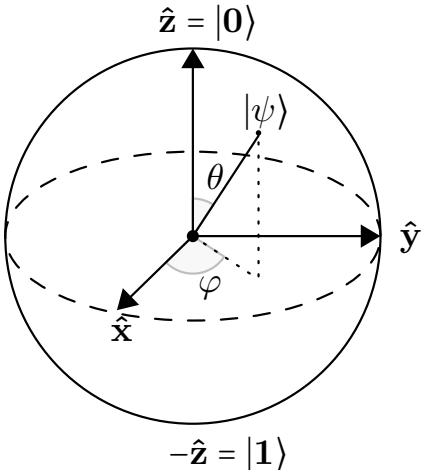
A quantum wave equation

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\Psi(\mathbf{r},t)$$

A Superposition of What?

In the abstract, a qu**b**it is a complex, linear superposition of two (exclusive and complete) states of a system in a two-dimensional Hilbert space — **not** *a vector in three dimensions*

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \alpha\begin{pmatrix}1\\0\end{pmatrix} + \beta\begin{pmatrix}0\\1\end{pmatrix}$$



Atomic (spin)

$$|\psi\rangle_{\text{electron-spin}} = \alpha\,|\uparrow\rangle_{\text{el}} + \beta\,|\downarrow\rangle_{\text{el}}$$
$$|\psi\rangle_{\text{nuclear-spin}} = \alpha\,|\uparrow\rangle_{\text{nuc}} + \beta\,|\downarrow\rangle_{\text{nuc}}$$

Photonic (polarization and number)

$$|\psi\rangle_{\text{photon-pol}} = \alpha\,|\nwarrow\rangle_{\text{ph}} + \beta\,|\nearrow\rangle_{\text{ph}}$$
$$|\psi\rangle_{\text{photon-num}} = \alpha\,|0\rangle_{\text{ph}} + \beta\,|1\rangle_{\text{ph}}$$

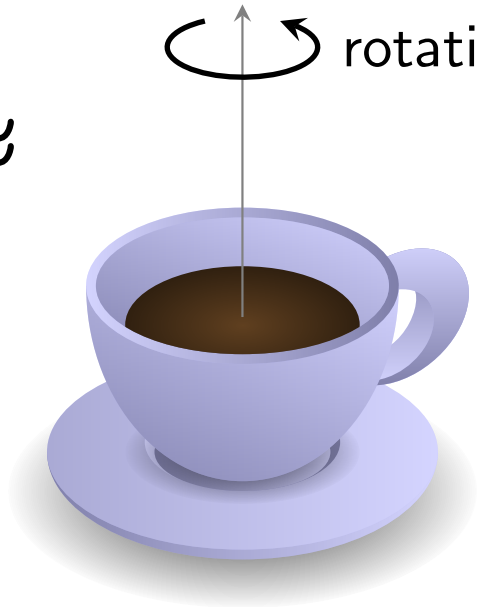
Electronic (charge, current)

$$|\psi\rangle_{\text{chg}} = \alpha\,|0\rangle_{\text{chg}} + \beta\,|2e\rangle_{\text{chg}}$$
$$|\psi\rangle_{\text{curr}} = \alpha\,|\circlearrowleft\rangle_{\text{curr}} + \beta\,|\circlearrowright\rangle_{\text{curr}}$$

Planck's Call to Action

The difference between classical physics and quantum physics is Planck's constant $\hbar$ (technically, $i\hbar$).

$\hbar \approx$



rotating a cup a billionth of a turn . . .

. . . over the lifetime of the universe!



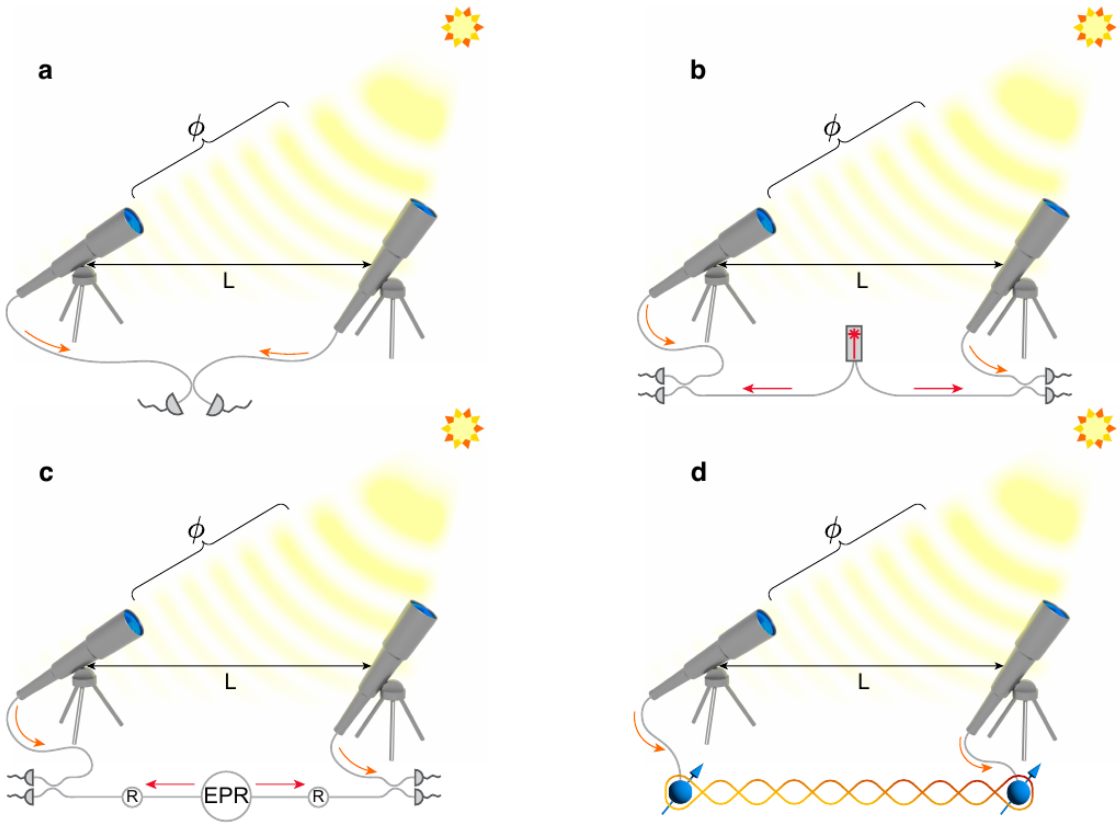
$\hbar \equiv 1.054\,571\,817 \times 10^{-34}\ \mathrm{J\,s}$

The Delicate, Competitive Dance of Quantum Information

- Isolation
  - Generated quantum states are fragile
  - Noise destroys, or at least complicates, their coherent nature
  - Segregation from environment and other systems is paramount . . .
- Interaction
  - . . . except splendid isolation leaves no means for control
  - Manipulation and detection of quantum states is necessary for utility
  - Quantum systems need to influence one another and another and . . .
- Interconnection
  - . . . a completely other
  - Collections of heterogeneous quantum systems must inter-communicate quantum information

Quantum Information for Very Long Baseline Interferometry (VLBI)

Four VLBI Schemes



Top row schemes are purely classical—(a) direct detection, (b) local oscillator-assisted– and the bottom are quantum-enhanced—(c) Gottesman-Jennewein-Croke (GJC), (d) Khabiboulline–Borregaard–de Greve–Lukin (KBGL) [11]

Recent Demonstration of KBGL Protocol

Entanglement Assisted Non-local Optical Interferometry in a Quantum Network

P.-J. Stas,[1,*] Y.-C. Wei,[1,*] M. Sirotin,[1,2,*] Y. Q. Huan,[1] U. Yazlar,[1,3]
F. Abdo Arias,[1] E. Knyazev,[1] G. Baranes,[1,2] B. Machielse,[4,5] S. Grandi,[6] D. Riedel,[5]
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(Dated: September 16, 2025)

arXiv:2509.09464

Enablers to Objectives

Quid nostra refert? (Of what importance is it to us?)

Enablers

- single photon light source
- **photon number resolving detector**
- **low-loss classical component**
- photonic integrated circuit

Applications

- quantum key distribution
- quantum secret sharing
- distributed quantum computing
- **distributed quantum sensing**

Technologies

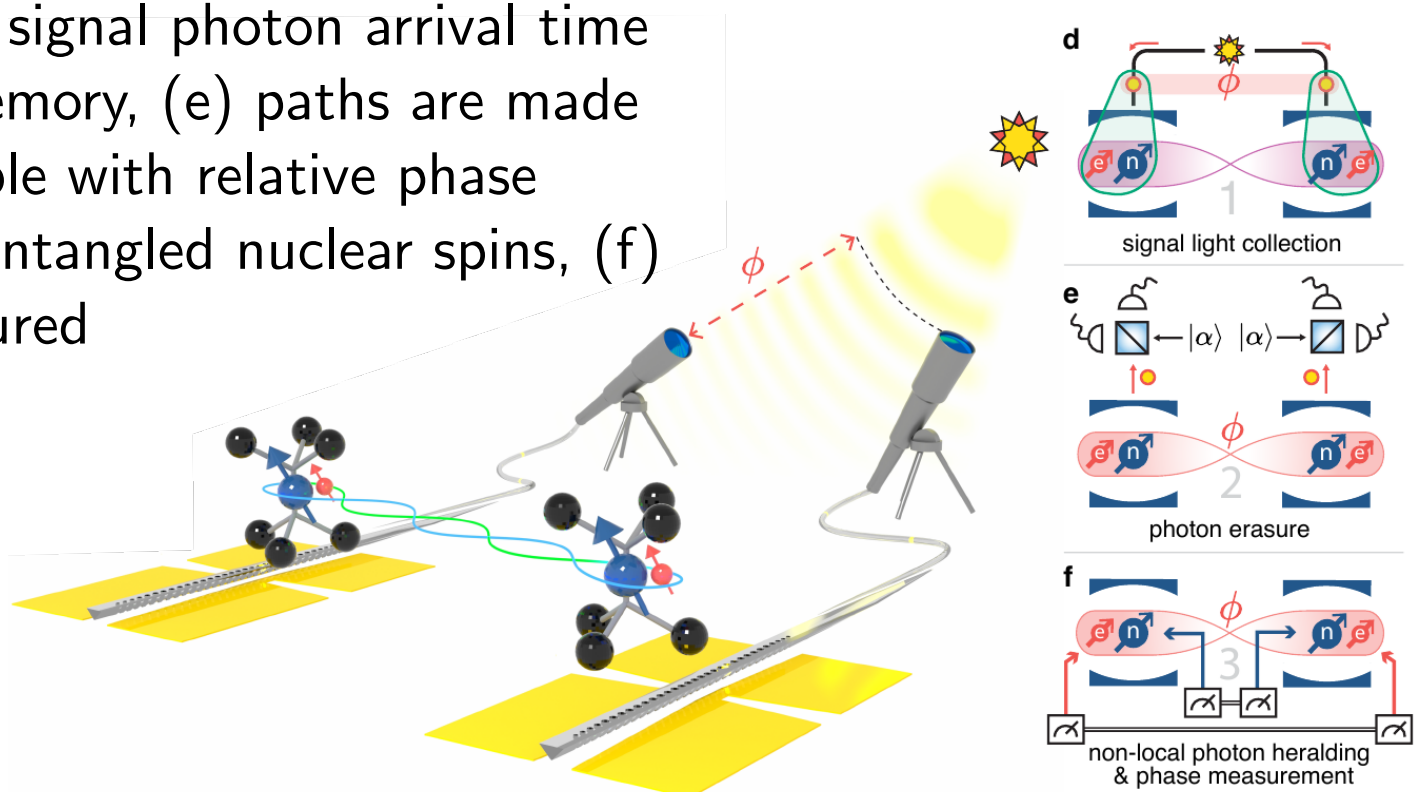
- **quantum memory**
- transduction
- quantum repeater
- **quantum processor**

Objectives

- contract non-repudiation
- chemical interaction modeling
- alternative GPS
- **astrophysical imaging**

KBGL VLBI Protocol Overview

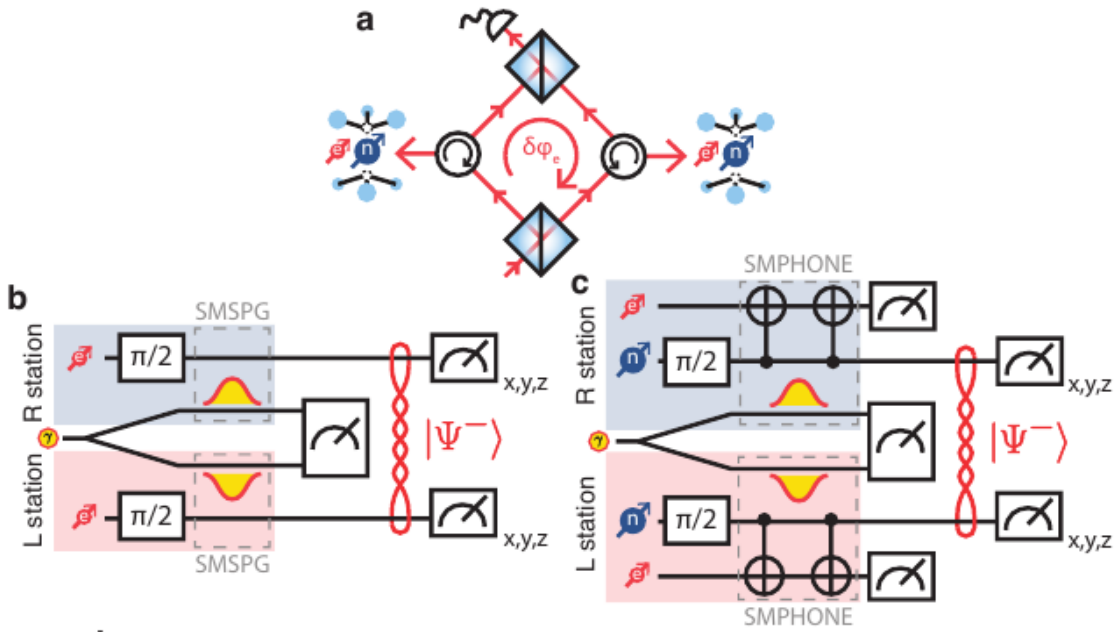
Matter memories at each collection site (Left and Right) are pre-entangled, (d) weak, thermal signal photon arrival time is stored in memory, (e) paths are made indistinguishable with relative phase imprinted on entangled nuclear spins, (f) phase is measured



KBGL VLBI scheme with pre-entangled memories [11]

Entangle Matter Memories

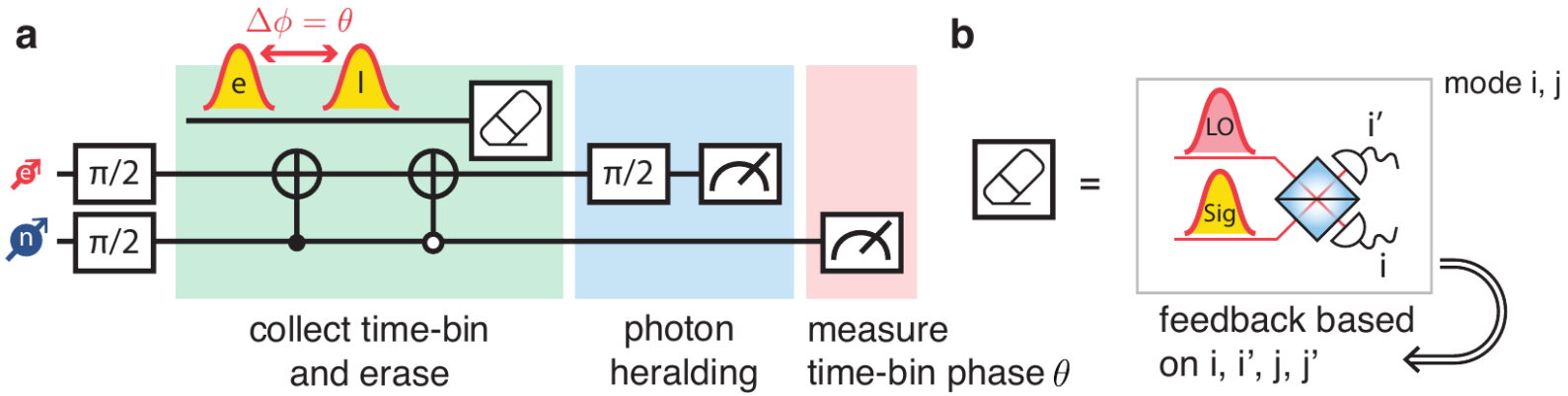
A reference pulse γ is shared between the matter memories of two stations via (a) Mach-Zehnder interferometer that (b) entangles the electron spins using single-mode spin-photon (SMSPG) gates, and subsequently (c) entangles the nuclear spins using single-mode photon-nucleus entanglement (SMPHONE) gates.



Entangle nuclear memories via electron spin entanglement [11]

Remove Which-Path Information: Erasure

Weak signal state emphasizing spatial separation:
$|\psi_{\text{sig}}\rangle \sim |0_L 0_R\rangle + \sqrt{\mu_{\text{sig}}/2}\left(|0_L 1_R\rangle + e^{i\phi}|1_L 0_R\rangle\right)$
Spatial separation yields temporal separation with a phase difference $\Delta\phi$.
Time bins are stored in the nuclear spin of the entangled memory using a binary encoding scheme with path information erased.
System state after erasure:
$|\psi_{\text{sys}}\rangle = |\downarrow\rangle_{\text{el}}|+\rangle_{\text{nuc}} + \sqrt{\mu_{\text{sig}}/2}\left[|+\rangle_{\text{el}}|+\rangle_{\text{nuc}} + e^{i\theta}\left(|+\rangle_{\text{el}}|\downarrow\rangle_{\text{nuc}} + |-\rangle_{\text{el}}|\uparrow\rangle_{\text{nuc}}\right)/\sqrt{2}\right]$
Apply Pauli Z operator $\hat{\sigma}_z$ on the nuclear spins to obtain relative phase θ.



Teleportation circuit (green part of a with b) with erasure operation [11]

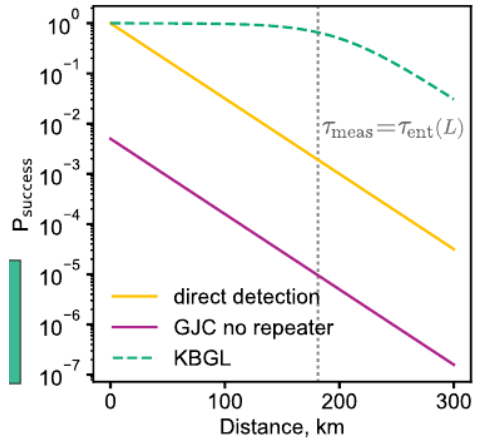
Measure of Quantum Advantage: Fisher Information

The Fisher information $\mathcal{F}_{\mathsf{I}}$ is identified as a signal-to-noise ratio $\varrho_{\mathsf{s:n}}$ metric, namely $\varrho_{\mathsf{s:n}} \equiv \sqrt{\mathcal{F}_{\mathsf{I}}}$.

A system will have its Fisher information reduced due to non-idealities, captured here in constants η_{erase}, η_{herald}, and $\bar{V}$. For the KBGL protocol, the Fisher information including imperfections is given by

$$\mathcal{F}_{\mathsf{I}} = \eta_{\mathsf{erase}}\eta_{\mathsf{herald}}\bar{V}^2\frac{\mu_{\mathsf{sig}}^2 e^{-2\mu_{\mathsf{sig}}}}{1-e^{-\mu_{\mathsf{sig}}}} \rightarrow \mathcal{F}_{\mathsf{I}} \propto \mu_{\mathsf{sig}} \text{ for } \mu_{\mathsf{sig}} \ll 1$$

which is the same scaling as classical direct detection.

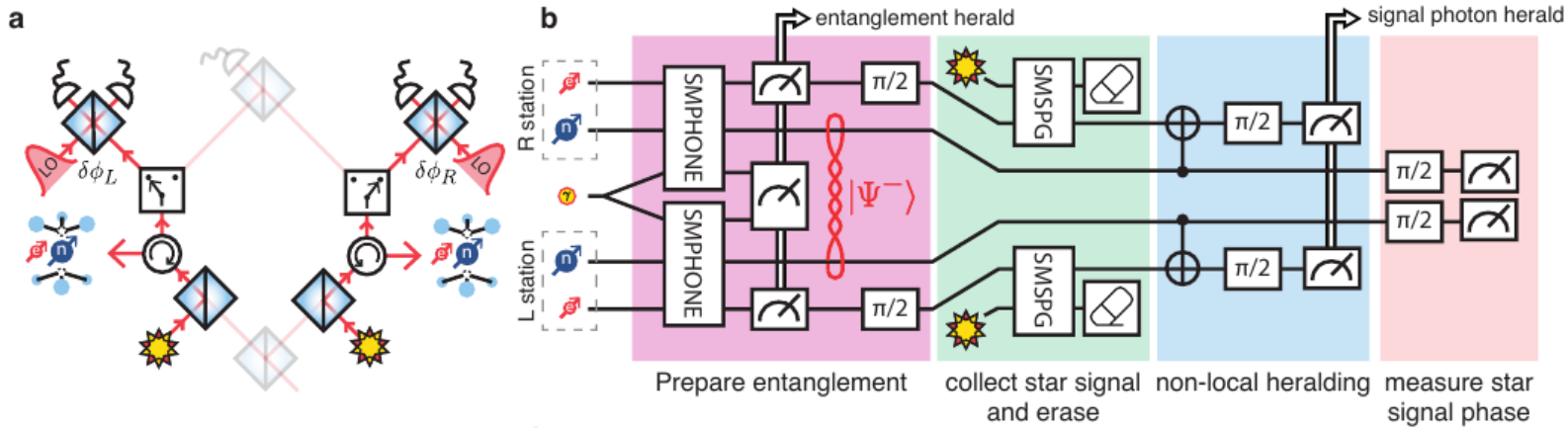


The KBGL protocol maintains significant distance advantage over direct detection [11]

KBGL VLBI Protocol Rearview

Key elements of the KBGL protocol:

- pre-entangled matter memories in cavities using MZI with local photons
- signal light time-bin arrival storage with path information erased for indistinguishability
- non-local heralding (not discussed) of signal photon to prepare for measurement
- phase measurement



Compact summary of KBGL protocol [11]

A Holy Curiosity

> *The important thing is not to stop questioning. Curiosity has its own reason for existence. One cannot help but be in awe when he contemplates the mysteries of eternity, of life, of the marvelous structure of reality.* **It is enough if one tries merely to comprehend only a little of this mystery every day.** *Never lose a holy curiosity.*

— Albert Einstein, *Death of a Genius* in LIFE magazine (April 1955)

Mysteries of Eternity, of Life, of [. . .] Reality



Milky Way over Lake Powell (Tim Burt, Nikon D7000, AF-S Nikkor 20mm, F1.8, 15s)

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[4] Jessica Illiano, Marcello Caleffi, Antonio Manzalini, and Angela Sara Cacciapuoti, *Quantum Internet Protocol Stack: a Comprehensive Survey*, (2022) https://arxiv.org/abs/2202.10894

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